Clustering and the Transfer-Operator Gap: A Machine-Checked Equivalence, and the Irrelevance of Observable-Dependent Constants

Lluis Eriksson

July 27, 2026 (version 1.0)

Abstract

Inside a Lean 4 formalization programme for four-dimensional $\mathrm{SU}(N_c)$ lattice Yang–Mills, we machine-check the step that the phrase *mass gap* names: the passage from exponential decay of a Euclidean correlator to a spectral gap of a transfer operator. For a symmetric bounded operator T on a Hilbert space with a fixed unit vector Ω, we prove that exponential decay at rate r of the *connected* two-point function $\langle v,\, T^n v\rangle - |\langle \Omega,\, v\rangle|^2$ is *equivalent* to the operator-norm bound $\|T - |\Omega\rangle\langle\Omega|\| \le r$; the conclusion is stated as a norm bound and not as an existential, and the equivalence is exhibited with inhabitants having a nonzero fluctuation sector. The sharp form is the substantive part. Writing $\mathcal{D}_r = \{v : \exists C,\ \forall n,\ \|S^n v\| \le C r^n\}$ for $S = T - |\Omega\rangle\langle\Omega|$, we prove that $\mathcal{D}_r$ is a *submodule* and that its density alone forces $\|S\| \le r$, with the constants entirely unconstrained: a family of observables whose span is dense, each carrying its own finite constant, suffices. Consequently a clustering estimate whose constant grows exponentially in the support of the observable — the shape produced by cluster expansions, and the shape of this programme's volume-uniform area law — is *not* an obstruction to the transfer-operator gap, and the uniformly quadratic hypothesis that the elementary proof requires is not necessary. We also record what the formalization does *not* contain: no Osterwalder–Seiler Hilbert space of states for any gauge theory, no reflection positivity of the Wilson measure, and no identification of a Euclidean correlator with a matrix element. Nothing here is a claim about the continuum limit or about the Clay problem. All results are machine-checked with no `sorry` and no project axioms; every headline is linked to its source and to its axiom-oracle transcript.

1 What is proved, and what is not

The gap in the chain

A lattice mass gap, in the sense of Osterwalder and Seiler [1], is a statement about the spectrum of an operator: the transfer operator T obtained from the Euclidean measure by reflection positivity satisfies $0 \le T \le 1$, fixes the vacuum vector Ω, and has a spectral gap on $\Omega^\perp$. Exponential decay of a Euclidean correlator is, by itself, a statement about a real-valued function of the separation. The two are connected by a theorem, not by terminology.

Within the formalization programme this paper belongs to, that theorem was absent. The terminal statement of the lattice assembly concludes about a function $\mathrm{cov} : \mathbb{N} \to \mathbb{R}$; the programme contains no Hilbert space of states, no transfer operator, no reflection positivity, and no spectrum. This paper supplies the *autonomous* half of the missing link: the operator-theoretic equivalence, at the level of generality where it is a theorem, machine-checked, together with the sharpening that determines what an application must supply.

Scope, stated in advance

The following are *not* established here, anywhere, in any form: reflection positivity of the Wilson gauge measure; the Gelfand–Naimark–Segal quotient; the existence of a transfer operator for any gauge theory; the identification $\mathbb{E}[A \cdot \theta_n B] = \langle A\Omega,\, T^n B\Omega\rangle$; density of any concrete family of gauge-invariant observables in the fluctuation sector of any such space. Consequently nothing in this paper is a claim about Yang–Mills, about the continuum limit, or about the Clay problem, and no statement below should be read as transferring importance from the subject matter of the programme to the theorems proved here. The equivalence itself is classical; see Section 8.

2 Setting

Let H be a Hilbert space over $\mathbb{R}$ (Sections 3) or over $\mathbb{C}$ (Sections 4–5).

Definition 2.1 (transfer data)**.** A pair (T, Ω) with T a bounded operator on H and $\Omega \in H$ is *transfer data* when T is symmetric, $\|\Omega\| = 1$, and $T\Omega = \Omega$. We write $P = |\Omega\rangle\langle\Omega|$ for the rank-one projection $Pv = \langle \Omega,\, v\rangle\, \Omega$ and

$$S \;=\; T - P$$

for the *vacuum-projected* operator. These are the properties that reflection positivity is used to produce; here they are hypotheses.

Definition 2.2 (connected two-point function)**.** For $v \in H$ and $n \in \mathbb{N}$,

$$G_v(n) \;=\; \langle v,\, T^n v\rangle \;-\; \langle v,\, \Omega\rangle\langle \Omega,\, v\rangle.$$

The elementary reason the *truncated* correlator, and not the full one, is the object that sees a gap is the following identity, which is proved by induction from $S\Omega = 0$ and $\langle \Omega,\, Sv\rangle = 0$.

Lemma 2.3 (structural identity)**.** *For transfer data and every $n \geq 0$, $S^{n+1}v = T^{n+1}v - \langle \Omega,\, v\rangle\, \Omega$, and hence $G_v(n+1) = \langle v,\, S^{n+1}v\rangle$.*

3 The equivalence

Theorem 3.1 (clustering $\iff$ gap)**.** *Let (T, Ω) be transfer data on a real Hilbert space and $r \geq 0$. Then*

$$\|S\| \;\leq\; r \qquad\iff\qquad \forall v \in H,\ \exists C,\ \forall n,\quad |G_v(n)| \;\leq\; C\, r^n.$$

Two features of the statement are deliberate. First, the conclusion is an operator-norm bound and not an assertion of the form $\exists\, m > 0$; the latter shape admits vacuous witnesses, and this programme's own history contains one. Second, the statement is an equivalence, so it cannot be weakened into vacuity in either direction.

Proof sketch. Forward: $|G_v(n)| = |\langle v,\, S^n v\rangle| \le \|v\|\,\|S^n\|\,\|v\| \le \|v\|^2 r^n$ for $n \ge 1$ by Lemma 2.3, and the $n = 0$ term is $\|v\|^2 - |\langle \Omega,\, v\rangle|^2 \in [0, \|v\|^2]$ by Cauchy–Schwarz. Backward: $\langle v,\, S^{2n} v\rangle = \|S^n v\|^2$, so clustering at even times gives $\|S^n v\| \le \sqrt{C_v}\, r^n$ pointwise; the Banach–Steinhaus theorem applied to the family $r^{-n} S^n$ upgrades this to $\|S^n\| \le M r^n$; and for symmetric S one has $\|S^2\| = \|S\|^2$ — proved from Cauchy–Schwarz, without the C^*-algebra API — hence $\|S\|^{2^k} \le M r^{2^k}$ for all k and $\|S\| \le r$. □

4 The sharp form: the constants do not matter

Theorem 3.1 assumes clustering at *every* vector. A cluster expansion never delivers that: it bounds a *family* of observables. On a merely dense set, the elementary argument — extend the quadratic-form bound by continuity — requires the constant to be uniformly quadratic, $C_v = K\|v\|^2$. That requirement is a property of that proof. It is not necessary.

Definition 4.1 (decay domain)**.** For a bounded operator S on H and $r \in \mathbb{R}$,

$$\mathcal{D}_r(S) \;=\; \big\{\, v \in H \;:\; \exists C,\ \forall n,\ \|S^n v\| \le C\, r^n \,\big\}.$$

Lemma 4.2. *$\mathcal{D}_r(S)$ is a submodule of H.*

Proof. $C_{u+v} \le C_u + C_v$ and $C_{\lambda v} = |\lambda|\, C_v$. □

Lemma 4.2 is the entire content of the sharpening: per-observable constants, however fast they grow along any exhaustion of the space, close up under linear combination with no uniformity whatsoever.

Theorem 4.3 (sharp form)**.** *Let S be a self-adjoint bounded operator on a complex Hilbert space $H \ne 0$ and $r \ge 0$. If $\mathcal{D}_r(S)$ is dense in H, then $\|S\| \le r$. Consequently, if $D \subseteq H$ has dense span and every $v \in D$ admits some finite C_v with $\|S^n v\| \le C_v r^n$ for all n, then $\|S\| \le r$.*

Proof. Suppose $\|S\| > r$. The norm is attained on the spectrum, so there is $\lambda_0 \in \sigma(S)$ with $|\lambda_0| = \|S\|$; choose $r < c < |\lambda_0|$ and set $f(x) = \max(0, |x| - c)$, so that f vanishes on $\{|x| \le c\}$ and $f(\lambda_0) > 0$. Define

$$k_n(x) \;=\; \frac{f(x)}{(\max(c, |x|))^{2n}}.$$

The denominator is $\ge c^{2n} > 0$ for *every* real x, so k_n is continuous with no case analysis and no division by zero, even when $0 \in \sigma(S)$; and $k_n(x)\, x^{2n} = f(x)$ identically, the even power letting $|x|$ stand in for x. Hence, in the continuous functional calculus, $f(S) = k_n(S)\, S^{2n}$, and on the spectrum $\|k_n\|_\infty \le \|S\|/c^{2n}$. For $v \in \mathcal{D}_r(S)$,

$$\|f(S)v\| \;=\; \|k_n(S)\, S^{2n} v\| \;\le\; \frac{\|S\|}{c^{2n}}\, C_v\, r^{2n} \;=\; \|S\|\, C_v \Big(\frac{r}{c}\Big)^{2n} \;\xrightarrow[n\to\infty]{}\; 0,$$

so $f(S)$ annihilates $\mathcal{D}_r(S)$, hence all of H by density and continuity. But $\|f(S)\| \ge |f(\lambda_0)| > 0$, a contradiction. The second statement follows from Lemma 4.2, which places the span of D inside $\mathcal{D}_r(S)$. □

Remark 4.4 (the result is not elementary)**.** *Three routes that suggest themselves all fail, and the reasons are worth recording. Banach–Steinhaus does not apply, because a dense* subspace *is meagre in general, so pointwise bounds on it yield no uniform bound. Powers do not suffice: $B = S^2/\|S\|^2$ is a positive contraction with $\|B\| = 1$ and $B^n v \to 0$ for every v — exactly what multiplication by x on $L^2[0,1]$ does — so no contradiction is available from norms alone. The resolvent route stalls: $v \in \mathcal{D}_r(S)$ lies in the range of $t - S$ for every $t > r$ by a Neumann series, but with a vector-dependent bound, and a self-adjoint operator with dense range need not be invertible. The statement is about the spectral measure near the top of the spectrum, and only a functional calculus sees that.*

Corollary 4.5 (support-exponential constants are not an obstruction)**.** *Let $\{A_s\}$ be observables whose span is dense in H, indexed so that A_s has support of size s, and suppose $\|S^n A_s\| \le C(s)\, r^n$ for all n, with a rate $r < 1$ independent of s. Then $\|S\| \le r$, whatever the growth of C. In particular $C(s) = e^{\kappa s}$ is admissible.*

Corollary 4.5 is the reason the sharp form matters for the programme this paper belongs to. The volume-uniform area law of [4] carries a constant of the form $N_c\, e^{|\mathrm{supp}|\cdot 4d\, K}$, exponential in the edge support of the loop, while its decay *rate* is the support-independent polymer activity rate; and the connecting-cluster estimate of the same programme has a support-independent constant but covers only a local, non-total family. A natural reading of that pair is that the two horns cannot be closed simultaneously, so that a total family must carry growing constants and the gap must fail volume-uniformly. Corollary 4.5 shows that reading is false: the growth of the constant is irrelevant, and only the uniformity of the *rate* is load-bearing.

5 The bridge in its own objects, and volume-uniformity

Theorem 4.3 is about a bare self-adjoint operator. Composing it with Definitions 2.1–2.2 gives the bridge in the form an application would consume.

Theorem 5.1 (sharp bridge)**.** *Let (T, Ω) be transfer data on a complex Hilbert space $H \neq 0$ with T self-adjoint, and let $r \ge 0$. Then $\|S\| \le r$ if and only if there is a set $D \subseteq H$ with dense span such that every $v \in D$ admits some finite C with $|G_v(n)| \le C\, r^n$ for all n.*

Theorem 5.2 (volume-uniformity)**.** *Let (T_i, Ω_i) be transfer data on complex Hilbert spaces $H_i \neq 0$, i ranging over any index set, and let $0 < r < 1$ be a* common *rate for which each i admits a family as in Theorem 5.1 — with per-observable constants that need not be related across i or within it. Then, with $m = -\log r > 0$,*

$$\|T_i - |\Omega_i\rangle\langle\Omega_i|\| \;\le\; e^{-m} \qquad \textit{for every } i.$$

Uniformity in i is not an added hypothesis: the conclusion of Theorem 5.1 is an operator-norm bound whose constant never sees the index. In the intended reading i is a finite volume and m is a volume-uniform mass; we stress that no transfer operator for any gauge theory is constructed here, so the reading is an intention and not a result.

6 Non-vacuity

Every statement above is exhibited with inhabitants in which the fluctuation sector is nonzero, so that none of them is satisfied only in the degenerate one-dimensional case.

Proposition 6.1. *Let Ω, w be orthonormal in H and $0 < r < 1$. Then $T = r\,\mathrm{id} + (1-r)P$ is transfer data with*

$$\|T - |\Omega\rangle\langle\Omega|\| = r \quad \textit{exactly}, \qquad T - |\Omega\rangle\langle\Omega| \neq 0, \qquad -\log r > 0.$$

The hypotheses are discharged concretely in $\mathbb{R}^2$ and in $\mathbb{C}^2$ with $\Omega = e_0$, $w = e_1$, so the family is inhabited.

7 What remains

With Corollary 4.5 in hand, the open content of the passage from Euclidean clustering to a lattice mass gap is the Osterwalder–Seiler construction and nothing else: reflection positivity of the Wilson gauge measure in the time direction; the GNS quotient and the transfer operator with $0 \leq T \leq 1$ and $T\Omega = \Omega$; the identification of the Euclidean correlator with a matrix element; and density, in the fluctuation sector of the resulting space, of a family of gauge-invariant observables for which a clustering estimate is available. Each of these is classical mathematics [1, 2] that has not, to our knowledge, been mechanized; none of them is established in this paper.

Remark 7.1 (a boundary of the formalization, not of the mathematics)**.** *Theorem 3.1 is formalized over a real Hilbert space and Theorems 4.3–5.2 over a complex one. The reason is that the library's continuous functional calculus for self-adjoint elements is derived from the complex case and is unavailable for operators on a real space, and the library contains no complexification of a real inner-product space. The two are separate instantiations of one argument; the transport between them is not formalized here and is not claimed. This is a boundary of the formalization, not of the mathematics.*

8 Reproducibility, verification links, and provenance

Freeze

All statements are machine-checked in Lean 4 (toolchain `leanprover/lean4:v4.29.0-rc6`) against Mathlib pinned to commit `07642720480157414db592fa85b626dafb71355b`. The freeze is the source tree at commit `ad9c93d793fa0b082f2373c2aea3ea5b99f9c6b8`; **`lake build YangMillsCore`** completes with **8415** jobs. Every declaration reachable from the core root is subject to an axiom oracle; the transcripts below record, for **2304 `#print axioms`** invocations, that the axiom sets occurring are exactly `[propext, Classical.choice, Quot.sound]`, `[propext, Quot.sound]` and `[propext]` — all subsets of the three permitted — with zero occurrences of `sorryAx` and no project axioms.

Theorem-to-artifact map

The three modules are `YangMills/OS/TransferGap.lean` (real case), `YangMills/OS/DenseClustering.lean` (sharp form) and `YangMills/OS/SharpBridge.lean` (sharp bridge); the column below names the module by initial.

Statement here	Lean name	Mod.
Def. 2.1 (P, S)	`vacuumProjection`, `projectedTransfer`	T

Statement here	Lean name	Mod.
Def. 2.2 (G_v)	`connCorr`	T
transfer data	`VacuumTransfer`	T
Lem. 2.3	`projected_pow_succ`, `connCorr_eq`	T
Thm. 3.1	`clustering_iff_gap`	T
$\|S^2\| = \|S\|^2$, proved by hand	`norm_sq_of_symm`	T
elementary dense form	`gap_of_dense_clustering`	T
Def. 4.1	`decayDomain`	D
Lem. 4.2	`decaySubmodule`	D
cut-off and factorisation	`cutOff`, `cutOffQuot`, `cutOffQuot_mul_pow`, `cutOffQuot_le`	D
Thm. 4.3	`norm_le_of_dense_decayDomain`, `norm_le_of_span_dense_decay`	D
Thm. 5.1	`sharp_clustering_iff_gap`	S
Thm. 5.2	`volumeUniform_sharp_gap` (and `volumeUniform_gap`, real case)	S, T
Prop. 6.1	`exists_vacuumTransfer_gap`, `euclidean_two_dim_vacuumTransfer_gap`	T
Prop. 6.1 (complex)	`exists_vacuumTransferC_gap`, `euclidean_two_dim_vacuumTransferC_gap`	S

Oracle transcripts and build root

- Core root: `YangMillsCore.lean`.
- Oracle driver: `oracle_check.lean`.
- Transcript at the real-case commit: `ORACLE-20260727-34696985.txt`.
- Transcript at the sharp-form commit: `ORACLE-20260727-a0ce50ea.txt`.
- Transcript at the sharp-bridge commit: `ORACLE-20260727-e5d76dae.txt`.
- Governance and scope record: `docs/O-BRIDGE-CHARTER.md` and `docs/O-BRIDGE-AUDIT-20260727.md`.

The transcripts were produced at the three code commits after which they are named; those commits were replayed over two intervening documentation-only commits before publication, and the replay is recorded together with the byte-level equality of the three module blobs.

Provenance and prior art

The equivalence between exponential clustering and a gap of the transfer operator is *classical* and no novelty is claimed for it: it is the standard argument of constructive field theory [2, 3], and the lattice-gauge transfer operator together with its positivity is due to Osterwalder and Seiler [1]. What is offered here is that the equivalence is machine-checked; that the dense-family form is stated with unrestricted constants, which is what isolates the uniformity of the rate as the load-bearing hypothesis; and Corollary 4.5. The volume-uniform area law whose constant shape motivates Corollary 4.5 is [4], the formal antecedent within the same programme.

References

[1] K. Osterwalder and E. Seiler, *Gauge field theories on a lattice*, Ann. Physics **110** (1978), 440–471.

[2] J. Glimm and A. Jaffe, *Quantum Physics: A Functional Integral Point of View*, 2nd ed., Springer, 1987; in particular §19.

[3] B. Simon, *The $P(\phi)_2$ Euclidean (Quantum) Field Theory*, Princeton University Press, 1974.

[4] L. Eriksson, *A volume-uniform area law for lattice Yang–Mills, machine-checked*, ai.viXra.org:2607.0005.

[5] L. de Moura and S. Ullrich, *The Lean 4 theorem prover and programming language*, CADE 28 (2021), 625–635.

[6] The mathlib Community, *The Lean mathematical library*, CPP 2020, 367–381.

[7] A. Jaffe and E. Witten, *Quantum Yang–Mills theory*, Clay Mathematics Institute Millennium Problem description, 2000.